B. Tech Examination-2022
Electronics and Communication Engineering
(Even Semester Regular and Supplementary)
Principle of Communication System (ECEUGOE02)

Full Marks: 80

Time: 3.00 Hrs

- ***Answer Q.1 and seven question from rest of the question.***
- *Answer all parts of a question in same place.*
- *Figures on the right-hand side margin indicate full marks.*
- *Symbols have their usual meaning*

1. Choose the correct alternative from the given options. 10

(a) The function of the input transducer in a communication system is – i) to transmit the message signal ii) to modulate the message signal iii) to convert message sound signal into electrical signal iv) none of the above

(b) If the radiated power of AM transmitter is 10 KW, the power in the carrier signal for modulation index of 0.6 is nearly – i) 8.24 KW ii) 8.47 KW iii) 9.26 KW iv) 9.6 KW

(c) A 1000 KHz carrier is simultaneously amplitude modulated with 300 Hz and 2 KHz audio sine wave. The frequency which will not present in the output is -i) 998 KHz ii) 999.7 KHz iii) 1000.3 KHz iv) 700 KHz

(d) In commercial FM broadcasting the maximum frequency deviation is normally -i) 5 KHz ii) 15 KHz iii) 75 KHz iv) 200 KHz

(e) Pre-emphasis in FM systems involves - a) compression of modulating signal b) expansion of modulating signal c) amplification of lower frequency components of the modulating signal d) amplification of higher frequency components of the modulating signal

(f) If a signal band-limited to f_m Hz is sampled at a rate less than $2f_m$ the reconstructed signal will be –i) higher in magnitude ii) smaller in magnitude iii) have higher frequency suppressed iv) distorted.

(g) Companding is used in PCM to – i) improve signal to noise ratio for low level input signal ii) reduce the probability of error iii) reduce quantizing noise iv) to increase signal strength.

(h) In a DM system the granular noise occurs when the modulating signal - i) increases rapidly ii) changes very slowly iii) decreases rapidly iv) the nature of modulating signal has nothing to do with the noise.

(i) How many bits would be required to represent 256 quantization levels in PCM? – i) 6 ii) 8 iii) 5 iv) 7.

(j) Which modulation technique is known as ON-OFF keying – i) ASK ii) FSK iii) PSK iv) DPSK

2. (a) Draw the block diagram of a communication system and briefly explain the function of each block. 5

(b) What is baseband transmission? 2

(c) Explain the needs of modulation in communication system. 3

3. (a) What is meant by amplitude modulation? 2

(b) Define the term modulation index for AM. 2

(c) Derive an expression for single tone amplitude modulated wave. 4

(d) Draw the waveform of over modulated AM wave and write the condition for over modulation. 2

4. (a) Describe the generation of AM signal using square law diode modulation technique 5

(b) How is a DSB-SC signal demodulated?

(c) Calculate the percentage of power saving for DSB-SC signal for percentage modulation of 50 %. 2

5. (a) What do you mean by frequency deviation and carrier swing? 4

(b) Derive an expression for single tone narrow band FM signal.

(c) A single tone frequency modulated wave is denoted by following expression $s(t) = 12\cos[6\times10^{8}t + 5\sin(1250t)]$. Determine carrier frequency, modulating frequency, modulation index and maximum frequency deviation 4 5

6. (a) Explain the varactor diode method for generation of FM signal. 2

(b) What are the drawbacks of direct method for FM generation? 3

(c) Explain Pre-Emphasis in FM signal generation. 2

7. (a) State sampling theorem. 2

(b) What is aliasing? How is it prevented? 4

(c) Explain ideal sampling technique.

(d) Find the Nyquist rate and the Nyquist interval for the signal $x(t)=\cos(4000\pi t)\cos(1000\pi t)$ 2 5

8. (a) Explain the operation of a PCM transmitter with suitable block diagram. 3

(b) Derive the expression for transmission bandwidth in a PCM system.

(c) A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that number of quantization level is 512. Determine transmission bandwidth and output signal to quantization noise ratio. 2 4

9. (a) Explain Delta modulation in detail with suitable block diagram.

(b) What are slope overload distortion and granular noise in delta modulation and how it is removed in ADM? 3

(c) Given a sine wave of frequency f_m and amplitude A_m applied to a delta modulator having a step size Δ. Show that slope overload distortion will occur if $A_m > \Delta/(2\pi f_m T_s)$. Here T_s is the sampling frequency 3 3

10. (a) What are the desirable properties of line code?

(b) The binary data 10101001 is transmitted over a baseband channel. Draw the waveform for the transmitted data using following format. i) Polar RZ ii) Split phase Manchester coding. iii) AMI iv) Polar Quaternary NRZ signaling. 4

(c) A digital data 10101101 is transmitted using digital modulation techniques. Draw the waveform of ASK signal, FSK signal, PSK signal corresponding to the given digital data. 3 2

11. (a) What do you mean by entropy?

(b) A discrete memoryless source (DMS) X has five possible outcomes with probabilities $P_1=1/2$, $P_2=1/4$, $P_3=1/8$, $P_4=1/16$, $P_5=1/16$. Find entropy and information rate if there are 16 outcomes per second. 4

(c) Determine the Huffman code for the following messages with their probabilities given. 4

X_1	X_2	X_3	X_4	X_5	X_6
0.3	0.25	0.2	0.12	0.08	0.05

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